MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE

EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPHL100 – INDUCTIVE LOGIC

Duration: 2 Hours Maximum Marks: 70

Part A

SHORT ANSWER

Answer any 8 questions

Each question carries 2 marks

1. Attribute affirmed or denied of the subject
2. A set of proposition – support conclusion
3. Conclusion logically follow from premises- sound argument -true proposition and valid proposition
4. Experiments test repeated observations- established law become theory
5. Combining two or more simple propositions
6. If two or more instances shares only one common factor
7. Removes irrelevant factors to isolate the true cause
8. Eating same food- people fall – common food is cause
9. Hypothesis- tentative assumption- theory- conformed hypothesis

10. General idea of thing
11. Reasoning from general to particular where conclusion necessarily follows
12. Lacks logical necessity- future may not resemble the past

Part B
SHORT ESSAY
Answer any 6 questions
Each question carries 5 marks

13. Informative , expressive,directive,ceremonial
14. Logical error
15. Provides factual information
16. Vague and un testable- lack empirical evidence- logically weak
17. Law of causation – every event must have a cause- uniformity of nature
18. Testing hypothesis through observation and experiment
19. Table of presence- table of absence-
20. Hypothesis from observation – steam force – experimentation – scientific theory
21. Based on habit- not logic- repeated experience creates expectation , not certainity
22. Deduction – started from general law- All men are mortal – Rama is a man- there fore Rama is mortal

Induction : started from particular proposition

Eg: crow 1 is black

Crow 2 is black

Crow 3 is black……

All crows are black

Part C
ESSAY
Answer any 2 questions
Each question carries 12 marks

23. Rules of correct reasoning- Deals with arguments- deductive and inductive logic
24. Elementary block of logic- express / truth false hood- simple compound- essential for arguments and reasoning
25. Law of identity- (A=A), Law of non-contradiction, law of excluded middle

26. Tentative explanation-verification- direct verification – indirect verification -confirmation – establishment -rejecting rival hypothesis